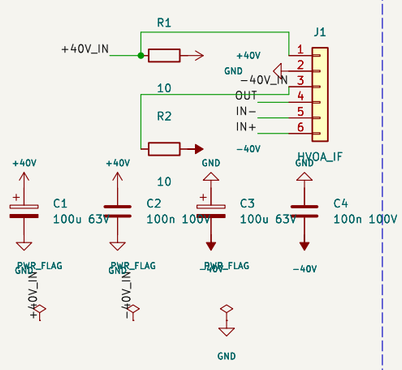
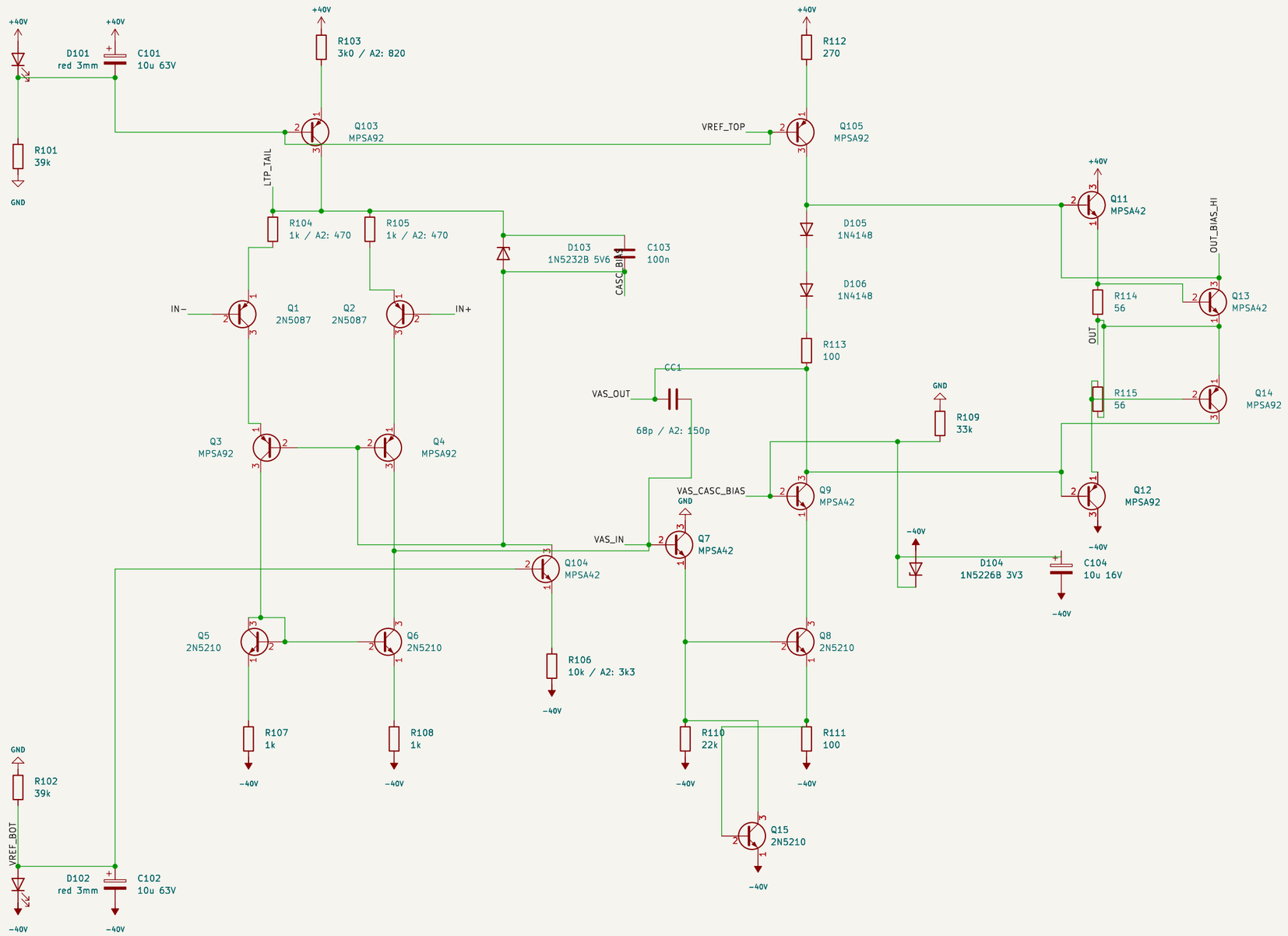
## Connector and rail filter




J1 plugs into the 6-pin socket on the preamp.

## HVOA-1 circuit




Q1/Q2 input pair runs at Vce ~4.7 V whatever the common mode: D103 bootstraps the Q3/Q4 cascode bases to TAIL.
Q9 cascodes the 2N5210 VAS (Vce ~2 V). Q13/Q14 limit output current to ~11 mA, Q15 limits the VAS to ~6 mA.

All TO-92 parts E-B-C (2N5087, 2N5210, MPSA42, MPSA92).
Match Q1/Q2 and Q5/Q6 and glue each pair together. D105/D106 touch Q11/Q12.
Topology identical to sim/hvoa.sub (checked by gen/build_all.py).
Build twice: A1 (instrument input) and A2 (FX return). Values marked "x / A2: y" differ.

Sheet: /
File: hvoa_module.kicad_sch

**Title: HVOA-1 +/-40 V nested-cascode discrete op-amp module**

| Size: A3 | Date: | Rev: 1 |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 1/1 |